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Carbon pass-through in Chinese cement industry

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ABSTRACT

This paper examines the carbon pass-through rate (CPTR) in Chinese cement industry using a rich dataset of daily data from the first trading day to December 2022. By using the Autoregressive distributed lag approach, we estimate the long-run effects of carbon costs on cement prices in five Chinese pilot carbon markets. We also control for the possible impact of the COVID-2019 epidemic and the de-capacity policies to ensure the robustness of empirical results. It has been found that downstream consumers bear more than half of the carbon emissions costs, with the magnitude of the pass-through rate differing significantly across regions. In particular, the CPTR in Beijing is almost 100 %, suggesting a high-level of windfall profits from free emission permits. Our study provides important implications for policy makers, including the need to consider reducing free allocation of permits for the cement industry given their high level of carbon pass-through, and provide different level of supports in different regions to balance the regional cement industry competitiveness.

1. Introduction

As a major policy instrument, the emissions trading system (ETS) has emerged as the most cost-effective way to achieve emission reductions in carbon-intensive sectors. The added carbon cost could reflect the change in the marginal cost of producers, which could affect the markets and consumers through the markup adjustment of product prices (Zhou et al., 2023). The carbon cost pass-through rate (CPTR), namely the changes in product prices caused by carbon cost, has increasingly become important in the field of industrial organization (Aguirre et al., 2010; Fabra and Reguant, 2014; Wang and Zhou, 2017). A higher pass-through rate indicates that consumers would bear most of the carbon cost, which could reduce product consumption and thus reduce carbon emissions. While a lower pass-through rate means that most of the carbon cost would be borne by producers, which may encourage investment in some carbon-efficient technologies. However, this may disadvantage the domestic carbon-intensive industries (Nelson et al., 2012; Nazifi, 2016; Wang and Zhou, 2017). Therefore, understanding and estimating CPTR can help regulators learn the cost burden borne by producers, and evaluate the need and scope of policy adjustments or industry assistance, including partial or complete allocation of free permits (Reinaud, 2007; Jouvet and Solier, 2013; Wild et al., 2015).

The existing studies on the CPTR are mainly concentrated on energy-intensive sectors covered by carbon markets across different regions. For example, in EU ETS, CPTRs are estimated in power sector (ranging from 74 % to 140 %) (Sijm et al., 2006; Reinaud, 2007; Kara et al., 2008; Fabra and Reguant, 2014; Dagoumas and Polemis, 2020; Pakrooh and Manera, 2024), cement sector (from 10 % to 75 %) (McKinsey and Ecofys, 2006; Demailly and Quirion, 2006; Walker, 2006; Alexeeva-Talebi, 2010; Vivid Economics and Ecofys, 2014; Cludius et al., 2020), paper sector (from 0 % to 50 %) (McKinsey and Ecofys, 2006; Alexeeva-Talebi, 2010), refining sector (from 25 % to 100 %) (Alexeeva-Talebi, 2010; Cludius et al., 2020), and steel sector (from 6 % to 66 %) (Alexeeva-Talebi, 2010; Cludius et al., 2020). Several scholars have analyzed the impact of Australian ETS on electricity prices, and the CPTRs are reported from 101 % to 393 % (Simshauser et al., 2007; Nazifi, 2016; Nazifi et al., 2021). Few studies have attempted to investigate the possible carbon pass-through in Chinese power sector, which shows no obvious linkage effect between carbon cost and power price (Wang et al., 2023).

As demonstrated in the review, the pass-through level typically depends on the industry category, market structure, price elasticity of demand, allocation rules and policy implementation (Sijm et al., 2012; Wang and Zhou, 2017; Pakrooh and Manera, 2024). Additionally, the different econometric models and data sets also lead to variation in the estimated CPTRs. While existing studies have made significant

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contributions to understanding CPTR in various sectors, a limitation lies in the temporal scope of many analyses. A substantial portion of the existing literature, particularly those focusing on the cement industry, has relied on relatively short-term datasets, potentially limiting the capacity to capture the long-term dynamics of CPTR. Moreover, there is few research in the literature regarding the CPTR in emerging carbon markets, especially in developing countries like China (Wang et al., 2023), where pilot carbon markets have been implemented for ten years.

China, as the world's largest producer of cement, accounted for approximately 57 % of the global production volume of cement in 2020 (Tan et al., 2022). Therefore, the carbon emission reduction in the cement industry is particularly important in China. Additionally, the carbon cost, as part of the production cost for cement enterprises, can directly influence their production decisions, emission reduction decisions and emission permit trading behavior (Yu et al., 2020). To mitigate the production burden of carbon costs on cement enterprises, they can pass through part or all of the costs to consumers by raising product prices. And the CPTR is an important indicator that reflects how the cement industry responds to the carbon market. The level of CPTR could directly affect the economic benefits and market competitiveness of cement enterprises. Therefore, it is necessary and meaningful to explore the CPTR of the cement industry in Chinese pilot carbon markets, and analyze its influencing factors and policy implications.

This paper contributes to estimate the CPTR of the cement industry in Chinese pilot carbon markets. We first apply the Autoregressive distributed lag (ARDL) model and use daily data to investigate the long-run effects of carbon costs on wholesale cement prices in 5 Chinese pilot carbon markets (including Beijing, Tianjin, Shanghai, Hubei, and Guangdong) from the first trading day to December 2022. Our results indicate that the opportunity cost of carbon emission is partly passed through by cement product pricing, and the CPTR estimation results vary from region to region. Besides, the over complete estimation results in Beijing suggest that companies can earn additional profits through carbon trading, which may be attributed to the market structure, allocation methods and other macro-economic factors. Meanwhile, we exclude the possible impact of the COVID-2019 epidemic and de-capacity policies, further verifying the conclusions. Besides, we apply the instrumental variable (IV) model and capture the effect of energy and weather variations, indicating the robustness of the results.

Our study could help scholars and regulators understand the impact of carbon markets on cement pricing. And we also provide important implications for policymakers and market participants in developing effective carbon markets. Specifically, our results suggest that the implementation of the ETS had a significant impact on cement prices, highlighting the need for the government to monitor CPTRs and adjust subsidy policies dynamically to maintain stable price levels. Moreover, we observe that de-capacity policies may affect CPTRs, which could have a positive effect on total emission reduction but not necessarily on overall emission intensity. Consequently, the design of national ETS should adjust the way of quota allocation method (especially the gradual transition from free quota allocation to auction allocation) to address the problem that companies can earn additional profits by carbon cost pass-through and facilitate carbon emission reduction in companies.

The remainder of this paper is organized as follows. Section 2 provides a detailed description of Chinese cement industry. Section 3 describes the variables, provides some preliminary data analysis, and introduces the Autoregressive distributed lag model to estimate the CPTR in Chinses cement industry. Section 4 presents the empirical findings. Meanwhile, several examinations are used to verify the results and test the robustness. The last section is the conclusions of this study.

2. The cement sector in China

Cement is fine material dust which has ideal physical properties such as heat insulation and viscosity and has been widely used as a basic building material (Madlool et al., 2011). Clinker is the main component of cement and is ground into cement together with other grinding materials. The key technique of clinker production is the calcination of limestone and silicate. The cement industry is energy-intensive, and the elementary sources of carbon emission are mainly from the productive process (Deja et al., 2010). It has been estimated that, on average, about 50 %–60 % of the carbon emissions for cement plants come from calcination process, and 40 % from fuel combustion. In addition, a proportion of carbon emissions, called indirect emissions (about 10 % of emissions from cement plants), is mainly from transportation and power utilization.

Since 1985, China has been the largest cement producer in the world, accounting for more than 50 % of the worldwide production. And the carbon emission from the cement industry have received attention from scholars and regulators (Hasanbeigi et al., 2013). As the economy grew, the cement output increased annually until 2014, when it peaked at 2.49 billion. With the weak performance of real estate and infrastructure in the downstream industry of cement demand, especially the decline in real estate investment, cement output in 2015 showed the first significant negative growth (−5 %) as shown in Fig. 1, with production remaining at 2.3 billion tons from 2015 to 2020. Meanwhile, with the innovation of production technology and the application of low-carbon equipment, the energy intensity and overall carbon emissions of the industry show a gradual decline, as shown in Fig. 2. Nevertheless, it is worth noting that the cement industry has overtaken the steel industry to become the second largest emitter (below the power industry), and it has been included in the seven pilot carbon markets in China. In 2022, about 2000 emitters and companies participated in the domestic pilot carbon markets, of which nearly 10 % were from the cement industry, specifically including 2 companies in Beijing, 10 in Tianjin, 7 in Shanghai, 45 in Hubei, 61 in Guangdong, and 33 in Chongqing.

As an important part of the cement industry index system, the Cement Price Index (CEMPI) could mainly reflect the comprehensive trend and extend of cement price changes in China.[1] As depicted in Fig. 3, the CEMPI has a huge range of changes, allowing us to determine the impact of the carbon markets and related policies on cement prices over time. According to our current statistics, cement prices within Chinese market decreased first and then risen as time went on. Specifically, in the first three years of the pilot markets establishment (2013–2016), the prices fluctuated and decreased significantly. While, the market price has shown a more stable trend with regular fluctuations since the gradually-stabilized emission standards were applied in the carbon market after 2017. During this period, more companies participated in the more strictly regulated carbon markets and were highly sensitive to the prices of certain administration policies. In terms of specific performance, the price increases are significant in autumn and spring, year after year. Moreover, the COVID-19 pandemic may have impact on the cement industry pricing. Actually, the COVID-19 pandemic has trigged the downturn in Chinese economy in the first several years. When the COVID-19 pandemic is controlled, there has been a surge in demand for raw materials as infrastructure projects resumed, leading to a substantial increase in cement prices in 2021.

In addition, some particular restriction policies for environmental protection, called de-capacity policies, have been implemented for producers to address the severe overproduction in the cement industry and to alleviate air pollution in winter (Zheng et al., 2019). In March 2014, the "two sessions" first put forward the policy expectation of production reduction requirements in the cement industry. In October 2016, the Ministry of Industry and Information Technology issued the "Notice on the Work of Cement Off-peak Production during 2016-2020 Rules" and made specific requirements on the production restrictions in

[1] The base year of CEMPI is 2009, and the base price is the national average price in 2009. The CEMPI is authorized by the Ministry of Industry and Information Technology (MIIT) and reported by China Cement network (https://www.ccement.com/)

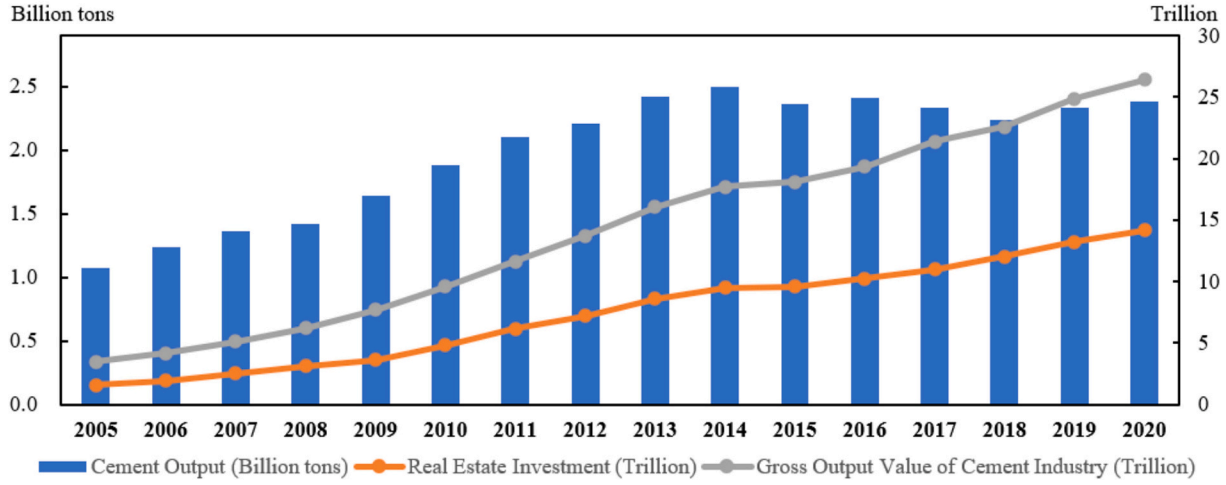


Fig. 1. The Real estate investment, gross output value of cement industry and cement output during 2005–2020.

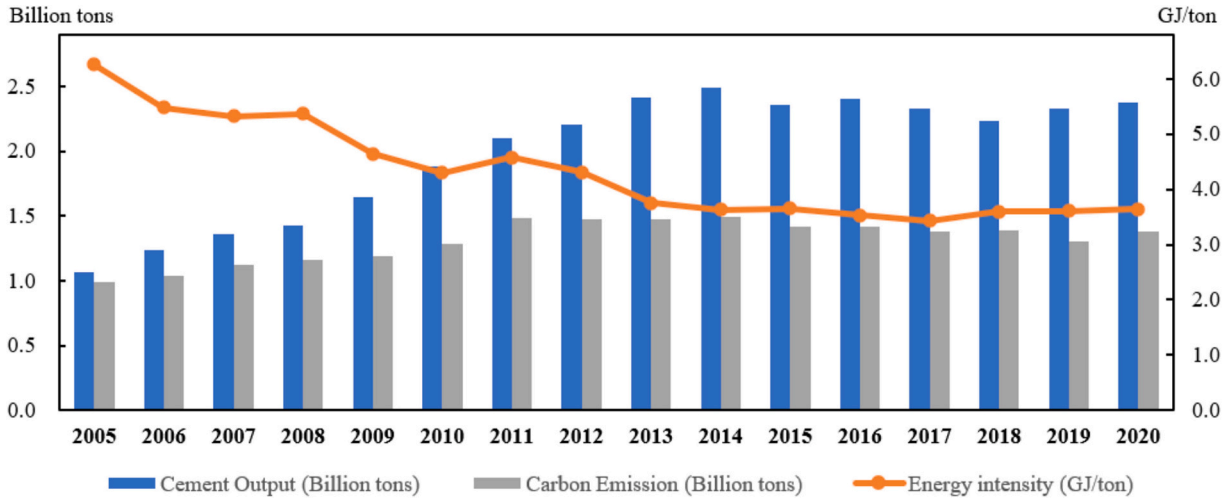


Fig. 2. The output, emission and energy intensity of Chinese cement industry during 2005–2020.

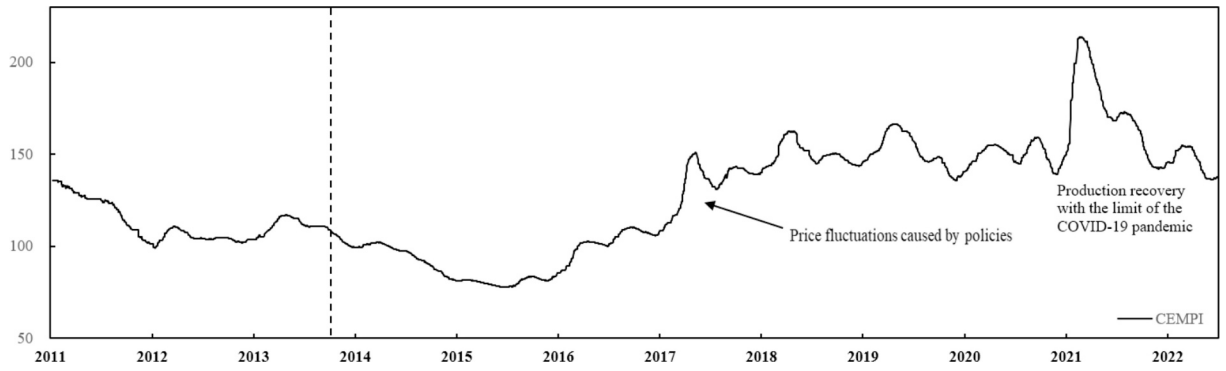


Fig. 3. The CEMPI of Chinese cement industry.

the cement industry. Since then, the relevant departments of state have discussed and formulated the production plans to promote the reduction strategies in the cement industry according to local conditions and explicitly implemented "differentiated" management based on the performance level of pollution emission. And in 2018, "The Notice of The State Council on the Issuance of the Three-year Action Plan for Winning the Blue-Sky Defense War" became a guiding document for the production of the cement industry in each region (Wang et al., 2021).

Cement companies in the areas with high air pollution must implement the de-capacity policies, which clearly limits operations in more areas. The relevant regulations take scientific and targeted measures to determine the duration of the de-capacity policies in different regions according to local conditions (e.g., the restriction period in Beijing, Tianjin, Hebei, and other places is from November 15 to March 15 of the

following year). In addition, kiln production is also banned during the rainy season in the south (for example, Jiangsu and Shanghai require a 15 day shutdown during the environmentally-sensitive period in summer, while companies in Hubei are required to stop production for no less than 20 days). As shown in Fig. 3, these policies had a certain impact on production, which may have caused fluctuations in the price of cement (Zhou et al., 2018).

As described by Syverson (2008) and Wang et al. (2021), the following characteristics of the cement industry provide ideal experimental conditions for this study. First, limestone, the core material in cement production (usually taking up to 70 % of the content), is costly to transport, leading to the cement plants being located near limestone mines. Second, high transportation costs and low industry profit margins limit the distance where products can be transported. These characteristics determine that the cement plants are usually located near the terminal consumption market. In China, the division of cement markets is almost consistent with the provincial administrative division (Wang et al., 2021). This regional feature effectively prevents transit of the markets that have never participated in the Emission Trading System to the markets that suffer an additional carbon cost under the ETS.[2] Finally, cement commodities are difficult to store because of their physical structure, which discourages cement companies from the strategic production behavior. In summary, the cement industry has become an ideal industry for economists investigating multiple topics (Syverson, 2008) and provides favorable preconditions for studying the carbon pass-through in our research.

3. Data and methodology

3.1. Data and variables

To empirically assess the impact of carbon emission costs on wholesale cement prices in cement markets from 2013 to 2022, this study utilizes regional-level data, and focuses on the wholesale cement prices, various fuel commodity prices, carbon emission prices and climate conditions by referring to the impact factors contained in relevant carbon pass-through literature (Sijm et al., 2006; Chen et al., 2008; Chernyavs'ka and Gullì, 2008; Nakamura and Zerom, 2010; Zhang et al., 2015).[3]

Data sources include specialized databases and platforms (Wind, CEIC and CSMAR), and government public documents which we have manually organized. Since the input price information is only available on weekdays, this paper fills in the weekend and holidays with the last available price data. In addition, this paper focuses on economic activity in six pilot markets, including Beijing, Tianjin, Shanghai, Hubei, Chongqing and Guangdong, and construct data analysis within these regions.[4]

The dataset is constructed daily from the first trading day to December 31, 2022. Chinese carbon pricing mechanism became effective in 2013, with Beijing and Shanghai opened carbon market in November 2013 at an initial price of 51.25 CNY/tCO_2 and 28.8 CNY/tCO_2, respectively. Guangdong and Tianjin also preliminarily completed their carbon market construction before 2014, with initial prices of 60.16 CNY/tCO_2 and 29.78 CNY/tCO_2. Subsequently, after a series of discussions and preparations, Hubei and Chongqing established the carbon emission trading system with initial price of 24.49 CNY/tCO_2 and 30.74 CNY/tCO_2, respectively.

As the marginal cost component, fuel prices were classified and derived for various products in different regions. To avoid the cost variance, we chose market prices of fuels in our study, which could better reflect the marginal cost of production. More specifically, we introduced the prices of Coal, Coke, and Brent, for estimation. Of which, Coal refers to the domestic thermal coal price; Coke is the price of industry coke used in production; while Brent refers to the spot price of crude oil. The fuel's system marginal prices of fuels are processed by the calorific values of different fuel types. In addition, actual values of temperature data from different regions are also employed in dataset and are adjusted to the underlying conditions of these pilot regions. In order to detailedly elaborate on the market price in each region, this paper also analyzes the actual values of large-scale producers, who have capability to meet the emission threshold requirements and have pricing-ability over years, as well as the average prices in regions. In Table 1, we provide the descriptive statistics about the sample variables of carbon and cement prices for all six regional markets and the period from the first trading day to 31 December 2022. As it is evident, the data show significant variability with the mean and the leptokurtic and platykurtic distributions derived from the results of Skewness and Kurtosis. As indicated by the mean values, the carbon prices of Beijing, Tianjin, Shanghai, Hubei, Guangdong and Chongqing are concentrated at 62.39, 20.57, 35.06, 27.19, 30.46, and 20.51 CNY/tCO_2. The standard deviation shows that the volatility of Guangdong and Beijing pilots is higher compared to other pilot markets.

We can get a more intuitive picture of the price series through Fig. 4, illustrating the significant variation in wholesale carbon and cement markets over the sample period. For all markets, daily prices are typically characterized by high standard deviation and show a strong seasonal component. However, this trend varies by region and is fully reflected in the carbon markets, where prices fluctuate frequently during the sampling period. For example, prices in Beijing usually rise in the summer and show a rapid decline in winter (see 2018 and 2020 in Fig. 4), which is related to the scale of local production and reflected in market transactions and demand. This also gives us an incentive to examine the changes in pass-through rates across regions empirically.

Throughout the development of the carbon markets, Beijing has maintained a high level of carbon emission prices, characterized by higher-frequency trading, with an average daily emission price of 62.35 CNY/tCO_2. This situation is associated with higher government emission reduction requirements, stricter carbon emission market system. However, the market prices in the pilot markets of Chongqing and Tianjin are relatively low, with an average price of 20.51 CNY/tCO_2 and 20.57 CNY/tCO_2, reflecting the low trading activity and imperfect market construction in these regions. In effect, the variation across the six pilot markets reflects that the carbon emissions prices are higher when carbon markets are relatively developed and more frequently traded. Based on our statistics, more tradings tend to be concentrated in the fulfillment period, reflecting the apparent performance-driving phenomenon in the pilot carbon markets. More specifically, the trading volume of these pilot carbon markets accounted for more than 50 % of the annual trading volume in May–July.

In most regions, there is typically a synchronous relationship between cement prices and carbon emissions, indicating the presence of carbon pass-through. However, this phenomenon does not seem to be present in Chongqing markets, which means there does not seem to be an obvious carbon pass-through. Notably, studies on China's pilot carbon markets frequently exclude the Chongqing ETS from sample data due to two primary reasons (Lyu et al., 2020). Firstly, the Chongqing ETS

[2] Although cement companies have been required by the government to move out of Beijing for better air condition since 2016, the production plants are mainly relocated in Hebei and Tianjin which is close to the Beijing wholesale market. We assume this case still satisfies the characteristics of regional market.

[3] This paper assumes that the main factors affecting cement prices can be predominately attributed to market demands, fuel costs, carbon prices and weather conditions. Therefore, the remaining factors which can influence wholesale prices (such as labor costs, return on capital for input, operational condition, depreciation expense and technology upgrade cost) are relatively constant over the empirical findings.

[4] Although we could access the price information of Shenzhen pilot carbon market from platform, the data of other variables in the estimation are prone to conflict with that of Guangdong. Hence, this paper does not analyze Shenzhen, which can be separately expanded in future research.

Table 1
Descriptive statistics for carbon prices (CNY/tCO_2) and cement prices (CNY/ton) in six regions from the first trading day to 31 December 2022.

Variables	Obs	Mean	Median	Min	Max	SD	Skewness	Kurtosis
Carbon Prices: CNY/tCO_2								
Beijing	3319	62.39	59.90	24.00	149.00	19.42	0.94	3.93
Tianjin	3300	20.57	22.50	7.00	62.38	8.33	0.27	2.34
Shanghai	3317	35.06	38.00	4.21	63.00	12.42	−0.38	3.46
Chongqing	2794	20.51	20.00	1.00	49.00	13.30	0.28	1.85
Guangdong	3307	30.46	23.05	1.27	95.26	21.10	1.21	3.23
Hubei	3170	27.19	25.54	9.38	61.89	10.01	0.74	2.90
Cement Prices: CNY/ton								
Beijing	3319	466.63	381	127	1019	215.80	0.84	3.25
Tianjin	3300	491.07	494	247	840	154.45	0.18	1.95
Shanghai	3317	404.80	461	150	634	128.41	−0.64	2.48
Chongqing	3300	369.40	408	221	711	101.81	0.29	2.51
Guangdong	3307	421.30	383	135	1005	201.26	0.67	2.43
Hubei	3170	381.46	402	200	623	91.27	0.18	2.01

has not operated efficiently, with trading volume approximately 1/15th of that of Guangdong's trading volume, rendering it an inadequate reflection of actual marginal emissions prices. Secondly, the Chongqing pilot market has a limited observation data duration, making it unsuitable for research purposes. Therefore, in the subsequent empirical analysis, we have excluded the Chongqing ETS from our sample data and focused on the carbon pass-through in other pilot markets.[5]

Although Fig. 4 provides evidence that the highly-volatile emission prices do affect the cement wholesale price to some extent, it does not tell us the extent of the impact, that is, the carbon pass-through rate. Therefore, the next matter of interest is to empirically analyze the CPTR. In the next stage of our work, it is necessary to conduct empirical studies by testing and digging into the dataset in-depth to guarantee the regression work appropriately.

3.2. Econometric model

In accordance with the fundamental principles of econometrics, it is essential that the techniques employed for estimating the parameters of the economic models satisfy certain criteria. Firstly, the time series data is subjected to routine preprocessing, and all variables are transformed into logarithmic form. Subsequently, standard methods are used to test the stationarity of economic series, and the data is adjusted and supplemented through normative operations where necessary. Specifically, the Augmented Dickey-Fuller (ADF) test, Phillips-Perron (PP) test, and Dickey-Fuller Test with GLS Detrending (DF-GLS) test are employed to assess whether the series follows the unit root, which is indicative of a non-stationary time series. The results of these tests are presented in Table B1, and the critical value for the unit root tests is set at 10 %.

We begin our modeling by referring to the most common used regression models used in previous studies (Zachmann and von Hirschhausen, 2008; de Bruyn et al., 2010; Alexeeva-Talebi, 2011; Cludius et al., 2020; Nazifi et al., 2021). Compared to other models, such as Ordinary least squares (OLS) model and Vector autoregression (VAR) model, the ARDL can model variables that are integrated of different orders if the Bound test (Pesaran et al., 2001) is satisfied. In addition, the ARDL model considers the different dynamic effects of the variables by setting different lag orders. Most importantly, the ARDL model is able to estimate the long-run and short-run impacts caused by changes in variables, the former of which is the research objective of this paper. We combine the ARDL model and form the following formula:

$$P_t = c_0 + c_1 t + \sum_{i=1}^{p} \varphi_i P_{t-i} + \sum_{i=0}^{q} \beta_i' P_{t-i}^c + \varepsilon_t \tag{1}$$

Where P_t is the wholesale price of cement markets at day t (common to all regions) and P_t^c denotes the cost of carbon permits required to cover emissions from generating a ton of cement. P_{t-i} and P_{t-i}^c reflects the influence brought by different lag periods, and $p \geq 1$, $q \geq 0$ is the lag order in the price series. c_0 is the intercept, reflecting some stable components. Finally, the impact of non-measurable components on the model is captured by the i.i.d (independent and identically distributed) error term, ε_t, where ε_t is normally distributed with zero mean and variance σ^2, i.e., $\varepsilon_t \sim N(0, \sigma^2)$. Rewriting the regression equation in error correction model by alternative error correction parameterization, we obtain the following Eq. (2).

$$\Delta P_t = c_0 + c_1 t - \alpha\left(P_{t-1} - \theta P_{t-1}^c\right) + \sum_{i=1}^{p-1} \Psi_{P_t} P_{t-i} + \omega' \Delta P_t^c + \sum_{i=1}^{q-1} \Psi_{P_t^c}{}' \Delta P_{t-i}^c + \mu_t \tag{2}$$

In this context, the speed of adjustment coefficient $\alpha = 1 - \sum_{j=1}^{p} \varphi_j$ and the long run coefficient $\theta = \frac{\sum_{j=0}^{q} \beta_j}{\alpha}$. And the coefficient θ represents the equilibrium CPTR, which indicates the percentage change in cement prices for a 1 % change in emission price in the long-term input cost function (Nakamura and Zerom, 2010; Guo and Gissey, 2021). Nazifi (2016) provides an explanation for the θ value, stating that if it equals 1, the entire cost of carbon emissions is reflected in the commodity price during production and sales, and companies pass through the total cost of carbon permits to consumers. If θ is greater than 1, it indicates a higher complete pass-through rate. From a microeconomics perspective, the commodity market may convert the carbon cost into a higher "mark-up" price due to the different conditions of supply and demand functions, resulting in the actual price exceeding the marginal production cost. When the θ is less than 1, it is clear that there is an incompleteness pass-through of the carbon cost (marginal production cost), indicating that some of the cost is absorbed internally by producing company. This situation may encourage the producers to invest in carbon reduction technologies, but it may affect the development of potentially intensive industries.

The reasoning behind this can be strongly illustrated by a number of event-driven factors, such as environmental-related events, demand change-related events, facility malfunction-related events or policy-related events. Due to lack of sufficient data or unmeasurable factors, it is difficult to fully assess the impact of all these events on cement companies (reflected in market segment). However, in this study, some policy-related events on spot prices are captured as a major factor in policy issuance and implementation in the cement industry through multiple regression analysis in the subsequent section.

[5] We also test the causal relationship between cement price and emission price in Chongqing pilot market, and the insignificant results are in line with our expectations. The results of Chongqing market are available from the authors upon request.

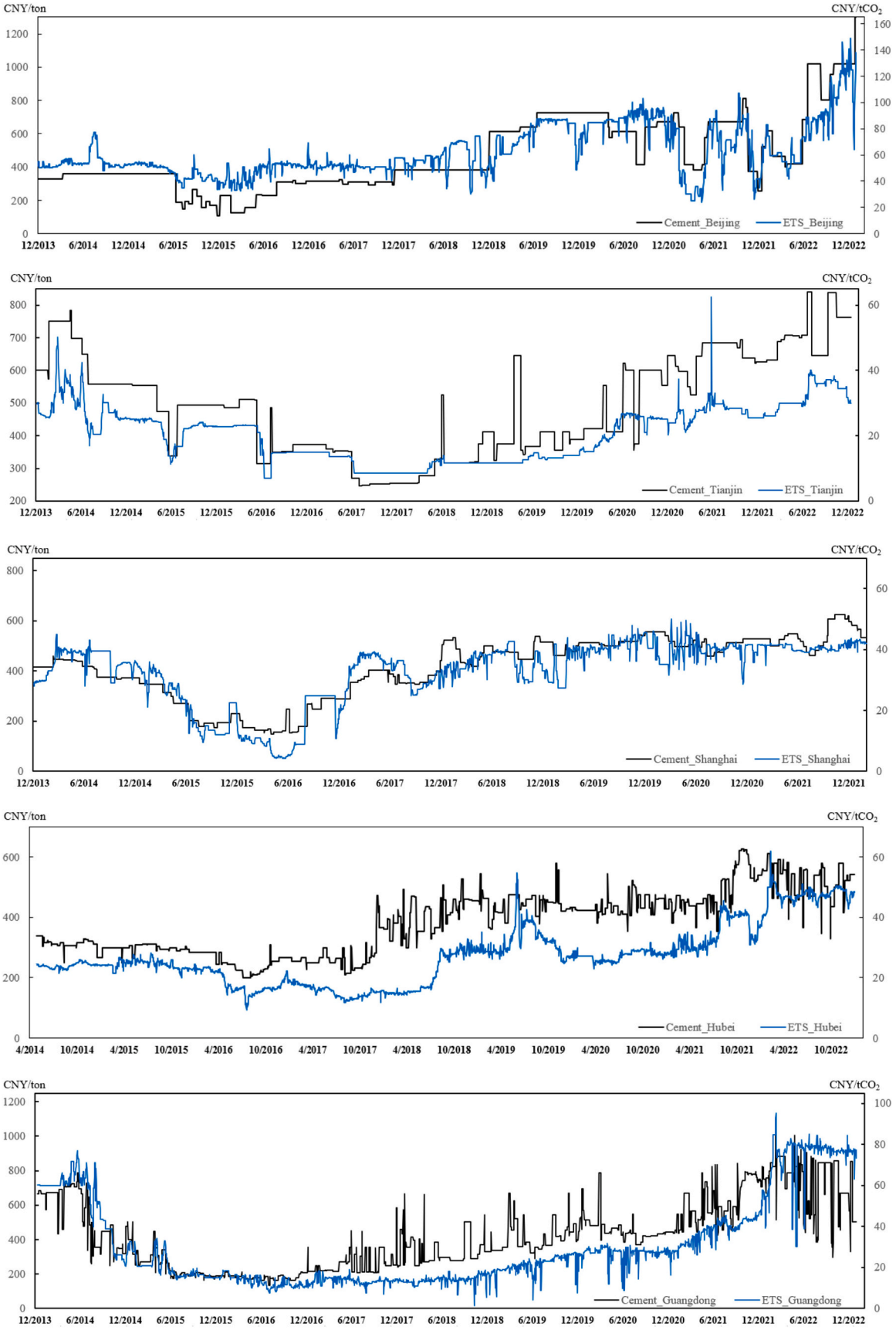


Fig. 4. Daily volume-weighted cement prices and carbon emission prices from the first trading day to December 30, 2022 in Beijing, Tianjin, Shanghai, Guangdong, Hubei, and Chongqing.

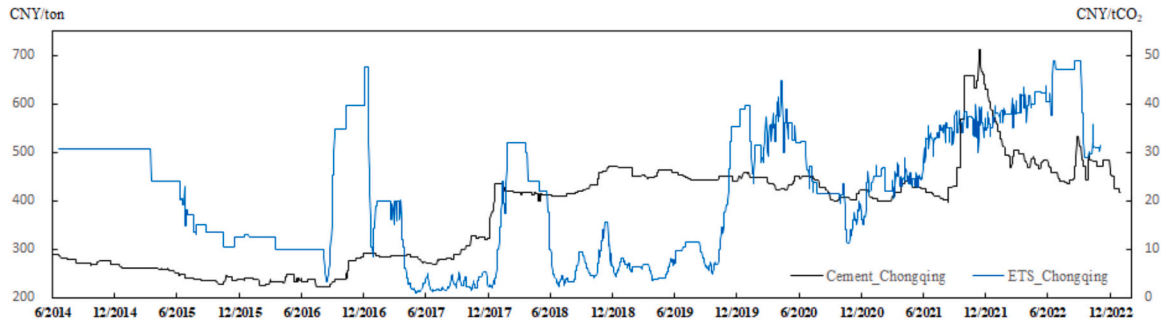


Fig. 4. (*continued*).

4. Results and discussions

4.1. Main results

Table 2 reports the empirical results of the carbon pass-through rates in cement markets by the ARDL model. As is evident, the bounds tests indicate a cointegration relationship for all pilot markets and the estimates for long-run CPTRs are statistically significant at the 1 % level. Both the statistical significance of CPTRs and test results support the assumption that the introduction of the carbon pricing mechanism promotes higher cement prices in the wholesale market. In addition, the results in Table 2 show that there are incomplete pass-through rates in

Table 2
CPTR regression results (ARDL model).

Dependent variable: ΔlnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
ΔlnCement (t-1)			−0.0261	−0.0681***	−0.2052***
			(−1.5193)	(−3.7760)	(−11.4320)
ΔlnCement (t-2)				−0.0702***	−0.0899***
				(−3.8960)	(−5.0042)
ΔlnCement (t-3)				−0.1011***	−0.0589***
				(−5.6261)	(−3.4133)
ΔlnCement (t-4)				−0.0506***	
				(−2.8337)	
ΔlnCement (t-5)				−0.0390**	
				(−2.1783)	
ΔlnCement (t-6)				−0.0397**	
				(−2.2297)	
ΔlnETS (t)	0.0122***	0.0281*	0.0063***	0.0875***	0.0546***
	(3.3144)	(1.7115)	(4.0459)	(2.8686)	(9.4136)
ΔlnETS (t-1)		−0.0148			
		(−0.8998)			
ΔlnETS (t-2)		0.0037			
		(0.2253)			
ΔlnETS (t-3)		−0.0063			
		(−0.3837)			
ΔlnETS (t-4)		−0.0549***			
		(−3.3645)			
EC estimations					
Long run pass-through	1.3890***	0.6675***	0.6849***	0.6191***	0.5399***
	(5.2625)	(13.0169)	(9.5681)	(11.8105)	(19.0511)
Speed-of-adjustment	−0.0088***	−0.0251***	−0.0093***	−0.0478***	−0.1011***
	(−3.7511)	(−5.7447)	(−4.2793)	(−7.0278)	(−10.7905)
Constant	0.0036	0.1052***	0.0337***	0.1883***	0.4253***
	(0.3697)	(5.7157)	(4.0130)	(6.7722)	(10.4822)
Diagnostics					
Observations	3311	3292	3369	3222	3359
Bound F-test	7.219**	16.600***	9.233***	24.713***	58.219***
Bound *t*-test	−3.751**	−5.745***	−4.279***	−7.028***	−10.790***
AIC	−11,975.09	−13,240.01	−16,660.75	−9783.768	−5512.597
BG (1)	0.323	0.103	2.163	1.024	1.159
BG (2)	0.379	0.863	2.468	3.060	3.355
BG (3)	0.488	1.048	2.666	4.878	4.664

Note(s): The sample period covers the first trading day to December 2022. *ln* stands for the natural logarithm. *Cement* and *ETS* refer to the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively. Lag orders for each ARDL model are selected on the basis of AIC with maximum lag orders of 8. Long run pass-through refers to the coefficient on θ in the cointegration equation. Speed-of-adjustment refers to the coefficient on α of the EC term. Both F- and *t*-tests examine long-run relationship discussed by Pesaran et al. (2001). AIC represents the Akaike information criterion. BG (1), BG (2), BG (3) are Breusch–Godfrey test statistics at lag order 1, 2, and 3, respectively. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

most pilot markets.

It is apparent from Table 2 that the estimated CPRTs vary from region to region and the estimated coefficients are 138.90 % for Beijing, 66.75 % for Tianjin, 68.49 % for Shanghai, 61.91 % for Hubei and 53.99 % for Guangdong. These differences between provinces could be attributed to different factory facilities (Sijm et al., 2006; Kim et al., 2010), production technology related to carbon intensity (Chernyavs'ka and Gullì, 2008; Wild et al., 2015; Nazifi, 2016), emission policies (Wang and Zhou, 2017) as well as the variances in market structure (Nakamura and Zerom, 2010; Fabra and Reguant, 2014). By calculating the speed-of-adjustment coefficient, it is shown that most of the changes in the price of carbon emission have a lagged effect on the wholesale cement market when estimating CPTRs. Most of the changes occur from over one week (Guangdong) to four months (Beijing) after the change in marginal cost, which is supported by Cludius et al. (2020) and Dagoumas and Polemis (2020).

The result reveals that in most local carbon markets, the increased costs brought by carbon emission are partly passed through to the wholesale markets, with both producers and consumers bearing the costs. In Guangdong, the long-run CPTR is the lowest (53.99 %), which may be attributed to the local companies' lower energy intensity. In particular, the manufacturers in the local market have access to advanced production processes and technologies (e.g., the pulverized coal burning equipment and exhaust gas treatment & burning equipment; clean-coal technology and CCS technology), which can help reduce production emissions. On the other hand, the estimated CPTR in Beijing is over 100 %, suggesting that companies in Beijing can earn additional profits through carbon trading. The primary factor contributing to the higher CPTR in Beijing is the market structure. Sijm et al. (2012) suggests that CPTR is determined by a combination of factors including the number of active companies in the market (N), the shape of the demand and supply curves.[6] They indicate that when N is small and demand is less elastic, the CPTR may significantly exceed 100 %. The number of cement companies participating in the carbon market in Beijing is significantly lower than other regions (only 2 companies), which implies a higher degree of market concentration. And it means that the Beijing cement market is dominated by a few large enterprises, which may have strong market power and can pass more carbon cost onto consumers. In contrast, there are many cement production companies in Guangdong and Hubei, and the market competition is relatively fierce, making it difficult for them to pass on the carbon cost to consumers.

It should be noted that the estimated CPTRs in our study are higher than those reported by several studies, which suggested relatively low CPTRs in the cement industry based on their empirical analysis. For instance, the estimated CPTRs in McKinsey and Ecofys (2006) range from 0 % to 15 %, in Walker (2006) are from 0 % to 50 %, and in Cludius et al. (2020) are from 2 % to 40 % in various EU member states. These differences in CPTRs may be attributed to variations in market structure and market maturity. The cement product trade between EU members, which may increase the competition in the cement market and reduce the market power of producers. Moreover, the carbon prices in EU ETS tend to be stable since the second stage, while the carbon prices in China pilot carbon markets show a larger price fluctuation due to a shorter operating time. In order to hedge against price risks, the domestic cement industry may tend to pass more carbon cost to consumers.

However, it should be emphasized that this long-term pass-through rate results ignore some positive impact of some short-term market behavior, which may be influenced by factors such as changes in production costs and conditions, seasonal factors and policies, and changes in market structure. Therefore, we apply more proper models considering more factors to further verify the estimation in the following robustness tests.

4.2. Robustness tests

4.2.1. Excluding the impact of COVID-19 effects

In this section, we conduct a series of examinations to improve the robustness of our results. Firstly, we exclude the influence of the COVID-19 pandemic on the estimation results. As we all know, 2020 was an exceptional year due to the worldwide health crisis caused by the COVID-19 pandemic, which had a substantial systemic impact across most sectors and led to significant losses in the industrial economy and industry development (Ren et al., 2022). The cement market supply and demand were also heavily impacted by the pandemic, which may ultimately affect the CPTR in the sector. For example, unprecedented delays, disruptions and uncertainty in construction projects could affect the market demand. Additionally, production restrictions, social distancing and quarantines requirements disrupted supply chains, causing a decrease in the market supply. Therefore, to ensure the reliability of our results, we exclude the data from 2020 to 2022, when the pandemic was at its height, and re-estimate the CPTR by the ARDL method.

On this basis, we estimate the CPTR excluding the impact of COVID-19 effects. The new results, reported in Table 3, demonstrate significant regional variations in CPTR, ranging from 50.16 % to 174.54 %. More specifically, the highest carbon pass-through occurred in Beijing (174.54 %), followed by Shanghai (67.04 %) and Tianjin (57.18 %), whereas the CPTR is lower in Hubei (54.03 %) and Guangdong (50.16 %). Overall, after considering the impact of COVID-19, the estimated CPTRs decreased in all regions except Beijing. Actually, the COVID-19 affected both the supply and demand of cement, leading to an increase or decrease in prices, as shown in Fig. 5. If the decrease in supply outpaced the decrease in demand, prices would rise; conversely, if the decrease in supply was smaller than the decrease in demand, prices would fall. The increased CPTR in Beijing may be attributed to a larger demand shock relative to supply shock caused by COVID-19, while the decreased CPTR in other regions suggests a smaller demand shock compared to the supply shock in those areas. Meanwhile, these results have important similarities with those reported in Table 2, indicating the robustness of our results across pilot markets.

4.2.2. Considering the de-capacity policy effects

In this section, we make a quantitative estimation of the impact of production policies and sharpen the robustness of our results. As described in Section 2, the Chinese government has implemented strict production policies (called de-capacity policies) for several years to control overproduction in various industries (Zhang et al., 2020; Wang et al., 2022). These de-capacity policies impose direct constraints on enterprises' production, leading to a reduction in market supply. Furthermore, de-capacity policies may cause the exit of high-cost or low-efficiency producers from the market, resulting in a more concentrated and scaled-up product market. Therefore, to assess the extent of the impact of de-capacity polies on the carbon pass-through, we construct an additional variable associated with the relevant policies and test the significance of this variable (Ren et al., 2022).

Table 4 shows the estimated coefficients of long-run CPTRs, considering the policy effects. All policy variables exhibit significant positive long-term coefficients, indicating the expected relationship between strict de-capacity policies and cement prices change. It can be seen that the estimated CPRTs are 92.51 % for Beijing, 71.24 % for Tianjin, 55.79 % for Hubei, 52.45 % for Guangdong and 50.86 % for Shanghai. All coefficients are statistically significant and have the expected signs. This suggests that the implementation of de-capacity policies had an impact on the assessment of CPTRs, and the estimations

[6] Sijm et al. (2012) provide the CPTR equation as $CPTR = \frac{1}{\left(1-\frac{1}{N\varepsilon}\right)(1+b\varepsilon)}$. N is the number of active companies in the market,1/b is the price elasticity of supply ($b > 0$), ε is the price elasticity of demand ($\varepsilon > 0$).

Table 3
CPTR regression results (Excluding the COVID effects).

Dependent variable: ΔlnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
Lagged terms (ΔlnCement, ΔlnETS)	Yes	Yes	Yes	Yes	Yes
EC estimations					
Long run pass-through	1.7454***	0.5718***	0.6704***	0.5403***	0.5016***
	(5.1052)	(7.2726)	(8.2115)	(5.4168)	(9.3911)
Speed-of-adjustment	−0.0105***	−0.0217***	−0.0111***	−0.0362***	−0.0803***
	(−3.1453)	(−4.0714)	(−3.8749)	(−4.9629)	(−6.9587)
Diagnostics					
Observations	2213	2202	2213	2066	2203
Bound F-test	4.951*	8.769***	7.692**	12.324***	24.476***
Bound t-test	−3.145*	−4.071***	−3.875***	−4.963***	−6.959***
AIC	−8639.191	−8791.032	−10,515.19	−6625.926	−3801.238
BG (1)	0.404	0.307	1.950	0.296	0.002
BG (2)	0.404	0.788	2.794	2.264	0.125
BG nnn	0.452	0.805	2.810	5.363	1.066

Note(s): The sample period covers the first trading day to December 2019. *ln* stands for the natural logarithm. *Lagged terms* include the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively, where lag orders for each ARDL model are selected on the basis of AIC with maximum lag orders of 8. Long run pass-through refers to the coefficient on θ in the cointegration equation. Speed-of-adjustment refers to the coefficient on α of the EC term. Both F- and *t*-tests examine long-run relationship discussed by Pesaran et al. (2001). AIC represents the Akaike information criterion. BG (1), BG (2), BG (3) are Breusch–Godfrey test statistics at lag order 1, 2, and 3, respectively. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

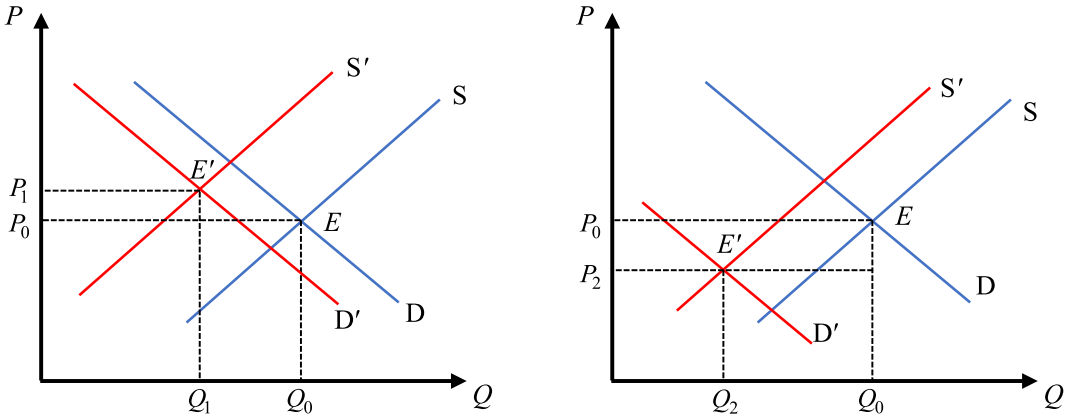


Fig. 5. Different effects of COVID-19 on the supply and demand of cement markets.

reported in Table 4 generally show lower results than the baseline model. This overestimation is mainly due to the fact that de-capacity policies could lead to higher prices, which is also reflected in Appendix C by using the SRD model.

For different regions, varying pass-through rates are measured, primarily influenced by the strength of policy implementation. For example, Beijing has the longest production suspension (approximately 150 days per year), which may have a more substantial impact on CPTR. Conversely, Hubei (60 days) and Guangdong (30 days) have shorter limits on production suspension, resulting in a smaller impact on estimation. It is worth noting that the CPTR of Tianjin reported in Table 4 is higher than in Table 2, which is contrary to other pilot markets. This difference may be related to other factors such as strict environmental protection policies, which may affect the demand of the cement product and hence the enterprises' pricing behaviors.

4.2.3. Instrumental variable model

We begin the CPTR measuring by the ARDL model. However, estimating Eq. (2) with OLS may be problematic due to the endogeneity issue. Previous research has shown that there is a correlation between carbon emission prices and the error term due to microeconomic reasons (Duso and Szücs, 2017; Dagoumas and Polemis, 2020). Therefore, considering this endogeneity issue, we apply the instrumental variable (IV) model and 2SLS approach. We use the lagged levels as the instrument of the endogenous variable (P_t^c) as suggested by Pancras et al. (2013) and Dagoumas and Polemis (2020). And the main specification of the instrumental variable (IV) model is as follows:

$$P_t = \alpha + \theta P_t^c + \beta_1 X_t^E + \beta_2 X_t^W + I_t + \varepsilon_t \tag{3}$$

The exogenous controls X_t^E and X_t^W represent energy and weather controls, respectively. X_t^E is a set of control variables that denote the wholesale prices of thermal coal (*Coal*), petroleum coke (*Coke*), and crude oil (*Brent*) in the commodity exchange market, respectively, while X_t^W represents a vector of exogenous control variables capturing the effect of climatic variations including regional average temperature and

Table 4
CPTR regression results (Considering policy effect).

Dependent variable: ΔlnCement)	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
Lagged terms (ΔlnCement, ΔlnETS, ΔPolicy)	Yes	Yes	Yes	Yes	Yes
EC estimations					
Long run pass-through	0.9251***	0.7124***	0.5086***	0.5579***	0.5245***
	(4.7953)	(15.0314)	(8.3711)	(16.0504)	(24.9300)
Policy	0.4550***	0.0777*	0.2857***	0.1996***	0.2031***
	(3.9074)	(1.6706)	(4.3270)	(6.9098)	(7.3549)
Speed-of-adjustment	−0.0142***	−0.0273***	−0.0138***	−0.0741***	−0.1399***
	(−4.8309)	(−6.4155)	(−5.2638)	(−8.9774)	(−12.5637)
Diagnostics					
Observations	3311	3292	3369	3222	3359
Bound F-test	7.941***	13.976***	9.404***	26.965***	52.735***
Bound t-test	−4.831***	−6.415***	−5.264***	−8.977***	−12.564***
AIC	−12,025.19	−13,523.35	−17,080.91	−9857.192	−5551.655
BG (1)	0.521	0.098	0.022	0.311	0.073
BG (2)	0.670	0.590	1.324	1.202	0.681
BG (3)	0.898	0.898	1.573	2.660	1.069

Note(s): The sample period covers the first trading day to December 2022. *ln* stands for the natural logarithm. *Lagged terms* include the wholesale cement price, emission price and de-capacity policies in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively, where lag orders for each ARDL model are selected on the basis of AIC with maximum lag orders of 8. Long run pass-through refers to the coefficient on θ in the cointegration equation. Speed-of-adjustment refers to the coefficient on α of the EC term. Both F- and *t*-tests examine long-run relationship discussed by Pesaran et al. (2001). AIC represents the Akaike information criterion. BG (1), BG (2), BG (3) are Breusch–Godfrey test statistics at lag order 1, 2, and 3, respectively. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

Table 5
CPTR regression results (IV model).

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
lnETS	0.9709***	0.6639***	0.4247***	0.4826***	0.4579***
	(64.4315)	(119.8826)	(64.5269)	(52.2617)	(47.6607)
Control variables	Yes	Yes	Yes	Yes	Yes
Diagnostics					
Observations	3318	3299	3376	3229	3366
Anderson LM statistic	3166.223***	3269.917***	3295.608***	3176.018***	2895.252***
Cragg-Donald Wald F statistic	68000***	360000***	140000***	190000***	20000***
Fixed effects (FE)					
Month FE	Yes	Yes	Yes	Yes	Yes
TEMP FE, highest temp	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	No	No
Month × TEMP, highest temp	No	No	No	No	No
Month × Year	No	No	No	No	No

Note(s): The sample period covers the first trading day to December 2022. *ln* stands for the natural logarithm. *Cement* and *ETS* refer to the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively. *Control variables* include *Coal*, *Coke*, *Brent*, *TempAver*, *TempHigh* are price of industry uses of thermal coal (CNY/ton), coke (CNY/ton), crude oil (CNY/ton), the average daily temperature (°C), and the maximum temperature (°C), respectively. Anderson LM statistic refers to the under identification test result for testing the rank of the IV model. Cragg-Donald Wald F statistic refers to the weak instrument test result for testing the relationship between the endogenous variables and the IVs. Sargan-Hansen test are unavailable to reported since models are just identified, i.e., the number of endogenous variables is equal to the number of IVs. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

maximum temperature. In this part, I_t is the vector of fixed effects, including the Month FE, Year FE and Temperature FE (highest temperature) to control for time-varying differences. The inclusion of these fixed effects in the instrumental variable regression enables a more accurate estimation of CPTRs.

The estimated coefficient of interest is reported in Row (2) of Table 5.[7] It can be seen that the estimated CPRTs are 97.09 % for Beijing, 66.39 % for Tianjin, 48.26 % for Hubei, 45.79 % for Guangdong and

[7] The more detailed CPTR regression results with more fixed effects are showed in Table B2-B4.

42.47 % for Shanghai. These results are consistent with the ARDL estimations reported in Table 4, indicating the robustness of the basic estimation.

5. Conclusions

This study empirically estimates the pass-through rates of carbon costs to wholesale prices in the cement market by applying ARDL models and further consider the COVID-19 and de-capacity effects in robustness tests. The results indicate that the opportunity cost of emission is passed through from companies to consumers with the estimated CPTRs are 138.90 % for Beijing, 66.75 % for Tianjin, 68.49 % for Shanghai, 61.91 % for Hubei and 53.99 % for Guangdong. As our results show, the magnitude of CPTRs varies across regions, which may be attributed to energy intensity, market structure, market demand, regional policies and other factors.

Based on the above research conclusions, some policy implications are proposed. First, more auctioning of carbon emission permits is suggested. Since cement produces pass through more than half carbon cost while they receive carbon permits for free, which leads to windfall profits and weakens the emission reduction incentives. By increasing the auctioning of emission permits, the government can increase the carbon cost for cement producers, thus may encourage them to adopt low-carbon technologies and practices. To facilitate the smooth transition, the government is strongly encouraged to implement a well-structured, multi-year permit allocation plan. A decadal or even longer timeframe is recommended to allow for a gradual and strategic reduction in the proportion of freely allocated permits, with the ultimate goal of their complete phase-out. By establishing such a clear and predictable transition pathway, enterprises will have a transparent and stable outlook. This will provide them with sufficient time and flexibility to carefully adjust their production and operations. Furthermore, it will enable them to make substantial investments in cutting-edge low-carbon technologies.

Second, to mitigate regional disparities in competitiveness, policymakers should consider providing targeted support to regions with higher carbon price transmission effects and weaker economic foundations, such as the differentiated allocation of free quotas. Specifically, more auctioning of emission permits is suggested in regions with higher CPTR to balance the regional differences in cement industry competitiveness, which could improve their energy efficiency and reduce emissions.

Despite the contribution of this paper, we use regional-level data to estimate CPTRs, which may not capture the heterogeneity and diversity of cement enterprises within each region. Therefore, more research is needed to use firm-level data to revel more information of the CPTRs among different types of cement enterprises, such as state-owned and private enterprises. This could help to understand the pricing decisions of cement enterprises more in detail and provide more policy implications for carbon market.

Nomenclature

Abbreviations

CPTR	Carbon pass-through rate
ETS	Emission trading system
ARDL	Autoregressive distributed lag model
IV	Instrumental variable model
SRD	Sharp Regression Discontinuity
LATE	Local Average Treatment Effect

Variables

ETS	The daily carbon prices in five pilot regionsCNY/tCO2
Cement	The daily cement prices in five pilot regionsCNY/ton
Coal	The domestic thermal coal priceCNY/ton
Coke	The price of industry coke used in productionCNY/ton
Brent	The spot price of Brent crude oilCNY/ton
TempAver	The average daily temperature°C
TempHigh	the maximum daily temperature°C

Parameters

P_t	The wholesale price of cement markets
P_t^c	The cost of carbon permits required to cover emissions
c_0	The intercept, reflecting the stable components
α	The speed of adjustment coefficient
θ	The long run coefficient
X_t^E	The energy controls
X_t^W	The weather controls
I_t	The vector of fixed effects
D_i	The dummy variable represents the implementation of the de-capacity policy

CRediT authorship contribution statement

S.Y. Liu: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **M. Wang:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Supervision, Validation, Writing – original draft, Writing – review & editing. **P. Zhou:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Supervision, Validation, Writing – original draft, Writing – review & editing.

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Appendix A. Appendix

Table B1
Unit root test.

Area	Variable	ADF statistic	PP statistic	DF-GLS statistic	Result
Beijing	lnCement	−3.373*	−3.451**	−2.852**	stationary
	lnETS	−4.160***	−5.344***	−3.900***	stationary
Tianjin	lnCement	−2.142	−2.871	−1.558	non-stationary
	ΔlnCement	−13.211***	−57.986***	−40.625***	stationary
	lnETS	−2.198	−2.254	−1.611	non-stationary
	ΔlnETS	−12.073***	−62.760***	−11.176***	stationary

(*continued on next page*)

Table B1 (*continued*)

Area	Variable	ADF statistic	PP statistic	DF-GLS statistic	Result
Shanghai	lnCement	−2.090	−2.338	−1.356	non-stationary
	ΔlnCement	−19.645***	−59.817***	−21.549***	stationary
	lnETS	−2.596	−2.776	−2.237	non-stationary
	ΔlnETS	−17.019***	−61.442***	−41.351***	stationary
Hubei	lnCement	−4.224***	−7.667***	−2.379	non-stationary
	ΔlnCement	−17.416***	−96.021***	−44.568***	stationary
	lnETS	−2.019	−2.427	−1.351	non-stationary
	ΔlnETS	−17.805***	−77.391***	−3.809***	stationary
Guangdong	lnCement	−3.509**	−7.692***	−1.527	non-stationary
	ΔlnCement	−15.437***	−144.998***	−50.888***	stationary
	lnETS	−2.130	−5.719***	−0.440	non-stationary
	ΔlnETS	−17.304***	−189.897***	−49.220***	stationary

Note(s): *ln* stands for the natural logarithm. *Cement* and *ETS* refer to the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively. Test equations include both trend and intercept terms for testing stationarity of ln*Cement* and *lnETS*, while include neither trend nor intercept term for testing stationarity of *ΔlnCement* and *ΔlnETS*. Lag orders for ADF test are selected by AIC. Newey–West lag orders for PP test are selected by Bartlett kernel spectral estimation method. Lag orders for DF-GLS test are selected on the basis of modified AIC. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

Table B2
CPTR regression results (IV model) with fixed effect (Year).

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
lnETS	0.6728***	0.5276***	0.2890***	0.3629***	0.3484***
	(33.3755)	(52.1109)	(54.6510)	(29.7561)	(17.6555)
Control variables	Yes	Yes	Yes	Yes	Yes
Diagnostics					
Observations	3318	3299	3376	3229	3366
Anderson LM statistic	2953.525***	3183.547***	3243.533***	2084.082***	1976.835***
Cragg-Donald Wald F statistic	26000***	89000***	81000***	67000***	4705.987***
Fixed effects (FE)					
Month FE	Yes	Yes	Yes	Yes	Yes
TEMP FE, highest temp	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Month × TEMP, highest temp	No	No	No	No	No
Month × Year	No	No	No	No	No

Note(s): The sample period covers the first trading day to December 2022. *ln* stands for the natural logarithm. *Cement* and *ETS* refer to the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively. *Control variables Coal*, *Coke*, *TempAver*, *TempHigh* are price of industry uses of thermal coal (CNY/ton), coke (CNY/ton), the average daily temperature (°C), and the maximum temperature (°C), respectively. Anderson LM statistic refers to the under identification test result for testing the rank of the IV model. Cragg-Donald Wald F statistic refers to the weak instrument test result for testing the relationship between the endogenous variables and the IVs. Sargan-Hansen test are unavailable to reported since models are just identified, i.e., the number of endogenous variables is equal to the number of IVs. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

Table B3
CPTR regression results (IV model) with fixed effect (Month × TEMP, highest temp).

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
lnETS	0.6655***	0.5211***	0.2917***	0.3621***	0.3531***
	(33.4245)	(51.5383)	(55.6162)	(29.4530)	(17.6005)
Control variables	Yes	Yes	Yes	Yes	Yes
Diagnostics					
Observations	3318	3299	3376	3229	3366
Anderson LM statistic	2947.918***	3194.165***	3240.499***	3080.503***	1947.532***
Cragg-Donald Wald F statistic	24000***	92000***	75000***	61000***	4319.402***
Fixed effects (FE)					
Month FE	Yes	Yes	Yes	Yes	Yes
TEMP FE, highest temp	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Month × TEMP, highest temp	Yes	Yes	Yes	Yes	Yes
Month × Year	No	No	No	No	No

Note(s): The sample period covers the first trading day to December 2022. *ln* stands for the natural logarithm. *Cement* and *ETS* refer to the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively. *Control variables Coal*, *Coke*, *TempAver*, *TempHigh* are price of industry uses of thermal coal (CNY/ton), coke (CNY/ton), the average daily temperature (°C), and the maximum temperature (°C), respectively. Anderson LM statistic refers to the under identification test result for testing the rank of the IV model. Cragg-Donald Wald F statistic refers to the weak instrument test result for testing the relationship between the endogenous variables and the IVs. Sargan-Hansen test are unavailable to reported since models are just identified, i.e., the number of endogenous variables is equal to the number of IVs. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

Table B4
CPTR regression results (IV model) with fixed effect (Month × Year).

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Variable					
lnETS	0.3049***	0.2560***	0.1150***	0.2252***	0.2065***
	(16.7240)	(11.6131)	(9.3908)	(6.4639)	(7.4486)
Control variables	Yes	Yes	Yes	Yes	Yes
Diagnostics					
Observations	3318	3299	3376	3229	3366
Anderson LM statistic	2394.724***	2379.314***	2213.219***	2094.237***	1121.871***
Cragg-Donald Wald F statistic	7682.609***	7606.055***	5763.449***	5305.896***	1527.736***
Fixed effects (FE)					
Month FE	Yes	Yes	Yes	Yes	Yes
TEMP FE, highest temp	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Month × TEMP, highest temp	Yes	Yes	Yes	Yes	Yes
Month × Year	Yes	Yes	Yes	Yes	Yes

Note(s): The sample period covers the first trading day to December 2022. *ln* stands for the natural logarithm. *Cement* and *ETS* refer to the wholesale cement price and emission price measured in Beijing, Tianjin, Shanghai, Hubei and Guangdong, respectively. *Control variables Coal*, *Coke*, *TempAver*, *TempHigh* are price of industry uses of thermal coal (CNY/ton), coke (CNY/ton), the average daily temperature (°C), and the maximum temperature (°C), respectively. Anderson LM statistic refers to the under identification test result for testing the rank of the IV model. Cragg-Donald Wald F statistic refers to the weak instrument test result for testing the relationship between the endogenous variables and the IVs. Sargan-Hansen test are unavailable to reported since models are just identified, i.e., the number of endogenous variables is equal to the number of IVs. t-statistics are reported in parentheses. ***, **, * indicate the statistic is significant at the 1 %, 5 %, and 10 % significance level, respectively.

Appendix B. Part of the research demonstrated that the implementation of the de-capacity policies increases the production prices in the pilot region, thus provides a good external perspective for our study. Considering that the de-capacity policies was rolled out to the whole country, we design the SRD model to test whether the implementation of the de-capacity policies raises the cement prices in the pilot regions, and the specific model can be expressed as follow

$$P_t = c_0 + \delta D_i + \beta f\left(x_t^p\right) + \gamma g\left(x_t^p \times D_i\right) + \alpha_1 X_t^E + \alpha_2 X_t^W + \varepsilon_t \tag{C1}$$

Where D_i is a dummy variable indicates the implementation of the de-capacity policy. x represents the number of days to the implementation date of the policy in each pilot region and the superscript p is its exponent from one to four. f denotes the function that sums the polynomials of x. δ is of interest to us, capturing the Local Average Treatment Effect (LATE) of de-capacity policy on cement prices. Table B1 presents the non-parametric estimation of SRD. The coefficients on LATE are positive and significant at the 1 % level regardless of the estimation conditions. Although the estimation of SRD does not directly measure the CPRTs, it verifies the consistency of the findings to a great extent.

Table C1
Estimation results for quartic polynomial SRD.

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Conventional	0.2724***	0.3022***	0.1957***	0.1075***	0.1857***
	(83.3303)	(16.4249)	(138.7598)	(5.1146)	(18.4190)
Bias-corrected	0.2741***	0.3060***	0.1954***	0.1125***	0.1929***
	(83.8456)	(16.6320)	(138.5573)	(5.3499)	(19.1314)
Robust	0.2741***	0.3060***	0.1954***	0.1125***	0.1929***
	(82.2854)	(16.0343)	(139.8494)	(5.3071)	(18.6530)
Polynomial order	4	4	4	4	4
Eff. Obs.	374	406	355	455	711
Left bandwidth	200.6317	113.7580	142.9029	174.9580	364.2201
Right bandwidth	173.1817	292.6359	212.8080	280.6025	346.7194

Note(s): Bandwidths on the left and right sides of the cutoff are optimally selected on the basis of minimizing the Mean-Square Error (MSE) of the local polynomial equation. Eff. Obs. refers to effective observations within bandwidths. Triangular kernel function is used to assign weights to observations within bandwidths. Conventional estimation refers to the Local Average Treatment Effect (LATE) without adjustment. Bias-corrected estimation uses Coverage Error-Rate (CER) optimal bandwidths for correcting the LATE. Robust estimation refers to the bias-corrected estimation with an asymptotic variance for the LATE. All parameters in the table are estimated based on a quartic polynomial SRD model.

Table C2
Estimation results for linear polynomial SRD.

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Conventional	0.2754***	0.3677***	0.1918***	0.1006***	0.2028***
	(105.2153)	(14.1909)	(517.7212)	(14.9478)	(32.2522)
Bias-corrected	0.2752***	0.3615***	0.1931***	0.0937***	0.1972***
	(105.1385)	(13.9500)	(521.0616)	(13.9302)	(31.3675)
Robust	0.2752***	0.3615***	0.1931***	0.0937***	0.1972***
	(102.5820)	(13.4892)	(411.6722)	(12.3294)	(25.7337)
Polynomial order	1	1	1	1	1
Eff. Obs.	101	70	116	248	174
Left bandwidth	25.7967	22.3969	59.9296	43.2660	167.3344
Right bandwidth	75.5256	47.1404	56.6519	204.5832	106.2144

Note(s): Bandwidths on the left and right sides of the cutoff are optimally selected on the basis of minimizing the Mean-Square Error (MSE) of the local polynomial equation. Eff. Obs. refers to effective observations within bandwidths. Triangular kernel function is used to assign weights to observations within bandwidths. Conventional estimation refers to the Local Average Treatment Effect (LATE) without adjustment. Bias-corrected estimation uses Coverage Error-Rate (CER) optimal bandwidths for correcting the LATE. Robust estimation refers to the bias-corrected estimation with an asymptotic variance for the LATE. All parameters in the table are estimated based on a linear polynomial SRD model.

Table C3
Estimation results for quadratic polynomial SRD.

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Conventional	0.2202***	0.2652***	0.2122***	0.0851***	0.1806***
	(85.8272)	(21.3951)	(178.8031)	(3.8455)	(21.2729)
Bias-corrected	0.2168***	0.2941***	0.2153***	0.0726***	0.1752***
	(84.5276)	(23.7229)	(181.4298)	(3.2806)	(20.6336)
Robust	0.2168***	0.2941***	0.2153***	0.0726***	0.1752***
	(78.5103)	(22.1740)	(132.9177)	(3.1891)	(18.5807)
Polynomial order	2	2	2	2	2
Eff. Obs.	241	277	204	238	342
Left bandwidth	176.0782	66.4825	128.5476	133.1535	141.4673
Right bandwidth	64.0533	210.7099	75.2496	104.0323	200.9421

Note(s): Bandwidths on the left and right sides of the cutoff are optimally selected on the basis of minimizing the Mean-Square Error (MSE) of the local polynomial equation. Eff. Obs. refers to effective observations within bandwidths. Triangular kernel function is used to assign weights to observations within bandwidths. Conventional estimation refers to the Local Average Treatment Effect (LATE) without adjustment. Bias-corrected estimation uses Coverage Error-Rate (CER) optimal bandwidths for correcting the LATE. Robust estimation refers to the bias-corrected estimation with an asymptotic variance for the LATE. All parameters in the table are estimated based on a quadratic polynomial SRD model.

Table C4
Estimation results for cubic polynomial SRD.

Dependent variable: lnCement	Beijing	Tianjin	Shanghai	Hubei	Guangdong
Conventional	0.2600***	0.2156***	0.2057***	0.1082***	0.1585***
	(76.5652)	(17.7316)	(140.9719)	(7.6176)	(17.0713)
Bias-corrected	0.2618***	0.2226***	0.1981***	0.1255***	0.1855***
	(77.0820)	(18.3028)	(135.7402)	(8.8313)	(19.9731)
Robust	0.2618***	0.2226***	0.1981***	0.1255***	0.1855***
	(76.1941)	(16.7027)	(110.4818)	(8.1453)	(18.6304)
Polynomial order	3	3	3	3	3
Eff. Obs.	292	429	345	391	454
Left bandwidth	104.6216	108.2873	198.0018	93.5979	190.3385
Right bandwidth	187.1771	320.0214	146.1639	297.6399	263.7072

Note(s): Bandwidths on the left and right sides of the cutoff are optimally selected on the basis of minimizing the Mean-Square Error (MSE) of the local polynomial equation. Eff. Obs. refers to effective observations within bandwidths. Triangular kernel function is used to assign weights to observations within bandwidths. Conventional estimation refers to the Local Average Treatment Effect (LATE) without adjustment. Bias-corrected estimation uses Coverage Error-Rate (CER) optimal bandwidths for correcting the LATE. Robust estimation refers to the bias-corrected estimation with an asymptotic variance for the LATE. All parameters in the table are estimated based on a cubic polynomial SRD model.

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eneco.2025.108413.

References

Aguirre, I., Cowan, S., Vickers, J., 2010. Monopoly price discrimination and demand curvature. Am. Econ. Rev. 100 (4), 1601–1615.

Alexeeva-Talebi, V., 2010. Cost PASS-through in Strategic Oligopoly: Sectoral Evidence for the EU ETS. Available at ftp://ftp.zew.de/pub/zew-docs/dp/dp10056.pdf.

Alexeeva-Talebi, V., 2011. Cost pass-through of the EU emissions allowances: examining the European petroleum markets. Energy Econ. 33, S75–S83.

Chen, Y., Sijm, J., Hobbs, B.F., Lise, W., 2008. Implications of CO_2 emissions trading for short-run electricity market outcomes in Northwest Europe. J. Regul. Econ. 34, 251–281.

Chernyavs'ka, L., Gullì, F., 2008. Marginal CO_2 cost pass-through under imperfect competition in power markets. Ecol. Econ. 68 (1–2), 408–421.

Cludius, J., Bruyn, S.D., Schumacher, K., Vergeer, R., 2020. Ex-post investigation of cost pass-through in the EU ETS—an analysis for six industry sectors. Energy Econ. 91, 104883.

Dagoumas, A.S., Polemis, M.L., 2020. Carbon pass-through in the electricity sector: an econometric analysis. Energy Econ. 86, 104621.

de Bruyn, S., Markowska, A., de Jong, F., Bles, M., 2010. Does the energy intensive industry obtain windfall profits through the EU ETS? An econometric analysis for products from the refineries, iron and steel and chemical sectors. Available at https://cedelft.eu/wp-content/uploads/sites/2/2021/04/7005_finalreportSdBEV.pdf.

Deja, J., Uliasz-Bochenczyk, A., Mokrzycki, E., 2010. CO_2 emissions from polish cement industry. Int. J. Greenhouse Gas Control 4 (4), 583–588.

Demailly, D., Quirion, P., 2006. CO_2 abatement, competitiveness and leakage in the European cement industry under the EU ETS: grandfathering versus output-based allocation. Clim. Pol. 6 (1), 93–113.

Duso, T., Szücs, F., 2017. Market power and heterogeneous pass-through in German electricity retail. Eur. Econ. Rev. 98, 354–372.

Fabra, N., Reguant, M., 2014. Pass-through of emissions costs in electricity markets. Am. Econ. Rev. 104 (9), 2872–2899.

Guo, B., Gissey, G., 2021. Cost pass-through in the British wholesale electricity market. Energy Econ. 105497.

Hasanbeigi, A., Morrow, W., Masanet, E., Sathaye, J., Xu, T., 2013. Energy efficiency improvement and CO_2 emission reduction opportunities in the cement industry in China. Energy Policy 57, 287–297.

Jouvet, P.A., Solier, B., 2013. An overview of CO_2 cost pass-through to electricity prices in Europe. Energy Policy 61, 1370–1376.

Kara, M., Syri, S., Lehtilä, A., Helynen, S., Kekkonen, V., Ruska, M., Forsström, J., 2008. The impacts of EU CO_2 emissions trading on electricity markets and electricity consumers in Finland. Energy Econ. 30 (2), 193–211.

Kim, W., Chattopadhyay, D., Park, J., 2010. Impact of carbon cost on wholesale electricity price: a note on price pass-through issues. Energy 35, 3341–3448.

Lyu, J., Cao, M., Wu, K., Li, H., Mohi-ud-din, G., 2020. Price volatility in the carbon market in China. J. Clean. Prod. 255, 120171.

Madlool, N.A., Saidur, R., Hossain, M.S., Rahim, N.A., 2011. A critical review on energy use and savings in the cement industries. Renew. Sust. Energ. Rev. 15 (4), 2042–2060.

McKinsey & Ecofys, 2006. EU ETS Review: Report on International Competitiveness. Available at http://ec.europa.eu/clima/policies/ets/docs/report_int_competitiveness_20061222_en.pdf. last accessed on 30 Oct 2014.

Nakamura, E., Zerom, D., 2010. Accounting for incomplete pass-through. Rev. Econ. Stud. 77, 1192–1230.

Nazifi, F., 2016. The pass-through rates of carbon costs on to electricity prices within the Australian National Electricity Market. Environ. Econ. Policy Stud. 18 (1), 41–62.

Nazifi, F., Truck, S., Zhu, L., 2021. Carbon pass-through rates on spot electricity prices in Australia. Energy Econ. 96, 105178.

Nelson, T., Kelley, S., Orton, F., 2012. A literature review of economic studies on carbon pricing and Australian wholesale electricity markets. Energy Policy 49, 217–224.

Pakrooh, P., Manera, M., 2024. Causality, connectedness, and volatility pass-through among energy-metal-stock-carbon markets: new evidence from the EU. Res. Policy 99, 105408.

Pancras, J., Gauri, D.K., Talukdar, D., 2013. Loss leaders and cross-category retailer pass-through: a Bayesian multileved analysis. J. Retail. 89 (2), 140–157.

Pesaran, M.H., Shin, Y., Smith, R.J., 2001. Bounds testing approaches to the analysis of level relationships. J. Appl. Econ. 16 (3), 289–326.

Reinaud, J., 2007. CO_2 Allowance and Electricity Price Interaction, Impact on industry's Electricity Purchasing Strategies in Europe. IEA Information Paper, International Energy Agency, OECD/IEA.

Ren, X., Duan, K., Tao, L., Shi, Y., Yan, C., 2022. Carbon prices forecasting in quantiles. Energy Econ. 108, 105862.

Sijm, J., Neuhoff, K., Chen, Y., 2006. CO_2 cost pass-through and windfall profits in the power sector. Clim. Pol. 6 (1), 49–72.

Sijm, J., Chen, Y., Hobbs, B.F., 2012. The impact of power market structure on CO2 cost pass-through to electricity prices under quantity competition–a theoretical approach. Energy Econ. 34, 1143–1152.

Simshauser, P., Doan, T., Lacey, B., 2007. The outlook for the economic and environmental performance of Australia's national electricity markets in 2030. Electr. J. 20 (6), 58–75.

Syverson, C., 2008. Markets: ready-mixed concrete. J. Econ. Perspect. 22 (1), 217–233.

Tan, C., Yu, X., Guan, Y., 2022. A technology-driven pathway to net-zero carbon emissions for China's cement industry. Appl. Energy 325, 119804.

Vivid Economics & Ecofys, 2014. Carbon Leakage Prospects Under Phase III EU ETS. Available at https://www.gov.uk/government/publications/carbon-leakage-prospects-under-phase-iii-of-the-eu-ets-and-beyond.

Walker, N., 2006. Concrete Evidence? An Empirical Approach to Quantify the Impact of EU Emissions Trading on Cement Industry Competitiveness. (Planning and Environmental Policy Research Series No. 06/10) (last accessed on 17 Nov 2014).

Wang, M., Zhou, P., 2017. Does emission permit allocation affect CO_2 cost pass-through? A theoretical analysis. Energy Econ. 66, 140–146.

Wang, Q., Xu, X., Liang, K., 2021. The impact of environmental regulation on firm performance: evidence from the Chinese cement industry. J. Environ. Manag. 299, 113596.

Wang, Y., Wang, D., Shi, X., 2022. Exploring the multidimensional effects of China's coal de-capacity policy: a regression discontinuity design. Res. Policy 75, 102504.

Wang, H., Feng, T., Zhong, C., 2023. Effectiveness of CO_2 cost pass-through to electricity prices under "electricity-carbon" market coupling in China. Energy 266, 126387.

Wild, P., Bell, W.P., Foster, J., 2015. Impact of carbon prices on wholesale electricity prices and carbon pass-through rates in the Australian national electricity market. Energy J. 36 (3), 137–153.

Yu, S., Fan, Y., Zhu, L., Eichhammer, W., 2020. Modeling the emission trading scheme from an agent-based perspective: system dynamics emerging from firms' coordination among abatement options. Eur. J. Oper. Res. 286, 1113–1128.

Zachmann, G., von Hirschhausen, C., 2008. First evidence of asymmetric cost of pass-through of EU emissions allowances: examining wholesale electricity prices in Germany. Econ. Lett. 99 (3), 465–469.

Zhang, Y., Wang, A., Tan, W., 2015. The impact of China's carbon allowance allocation rules on the product prices and emission reduction behaviors of ETS-covered enterprises. Energy Policy 86, 176–185.

Zhang, W., Meng, J., Tian, X., 2020. Does de-capacity policy enhance the total factor productivity of China's coal companies? A regression discontinuity design. Res. Policy 68, 101741.

Zheng, J., Mi, Z., Coffman, D., Milcheva, S., Shan, Y., Guan, D., Wang, S., 2019. Regional development and carbon emissions in China. Energy Econ. 81, 25–36.

Zhou, S., Wang, Y., Yuan, Z., Ou, X., 2018. Peak energy consumption and CO_2 emissions in China's industrial sector. Energ. Strat. Rev. 20, 113–123.

Zhou, P., Wang, J., Wang, M., 2023. The micro-effect of carbon emission trading policy: perspectives of firm green innovation and heterogeneity. J. China Univer. Petrol. (Edit. Soc. Sci.) 5, 19–33.